

**VoltraBloom: Conceptual Design of a Portable Hybrid Renewable Emergency
Power System Integrating Solar, Wind, and Soil Energy**



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Abbreviations: HEMS – Hybrid Energy Management System; MPPT – Maximum Power Point Tracking; SMFC – Soil Microbial Fuel Cell; IoT – Internet of Things; DoD – Depth of Discharge; ATS – Automatic Transfer Switch.

ABSTRACT

This study aims to design VoltraBloom, a portable hybrid renewable power system developed at an implementation scale to supply electricity for a dormitory or small building, while also serving as a backup power source for the wider community. The background of this research is the limited electrical capacity in dormitory buildings, which often causes frequent power interruptions, as well as dependence on diesel generators that are expensive and environmentally unfriendly. The method used is Conceptual Design Research (Research by Design). At the implementation scale, the study focuses on system design without physical fabrication or full-scale testing; however, a small physical proof-of-concept has since been built to verify the basic architecture of the design before it is scaled up.

VoltraBloom, at implementation scale, integrates a solar panel array of approximately 3.2 kWp, a vertical-axis wind turbine rated at 800 W, and a Soil Microbial Fuel Cell (SMFC) as an additional energy source. These are combined with a 48V 200Ah LiFePO₄ battery bank, a 48V/80A MPPT charge controller, a Hybrid Energy Management System (HEMS) based on ESP32, and a 5 kW pure sine wave inverter that delivers 220 V AC to the building distribution panel. The system is also equipped with an IoT-based monitoring system to track energy conditions in real time. These figures remain the implementation-scale target: the physical proof-of-concept currently uses four selected solar panels, a DC-motor wind-generation demonstrator, a simplified two-cell (2S) storage configuration, an Arduino controller, and does not yet reproduce the full-scale ratings above.

The research outputs show the potential of VoltraBloom as a portable and eco-friendly hybrid backup power system at the building scale, now supported by an initial physical proof of concept that demonstrates the intended relationship between renewable energy input, conditioning/storage, control, and mechanical deployment.

Keywords: VoltraBloom, renewable energy, hybrid system, building backup power, IoT monitoring, soil microbial fuel cell, proof of concept.

CHAPTER I

INTRODUCTION

1.1 Background

Electricity, according to the U.S. Energy Information Administration (EIA, 2025), is the flow of charge and is one of the most fundamental forms of energy in everyday life. The alternating current (AC) system, pioneered by Nikola Tesla and commercialised by George Westinghouse at the end of the 19th century, began replacing the earlier direct current system.

In practice, however, this condition is not always achieved. In one school dormitory environment, a setting marked by continuous residential and academic energy demand (Manaf, 2022), the electricity supply often experiences sudden outages. These outages occur unpredictably, morning, afternoon, or night, indicating that the allocated power capacity for the building is insufficient to support daily operations reliably, thereby causing repeated disruptions to learning activities and the use of basic electronic devices.

Ideally, this issue could be resolved by increasing the building's electrical network capacity. However, such a solution requires comprehensive remapping and large-scale installation upgrades, which are complex and costly to implement in a school environment. A more realistic alternative is a building-scale independent backup power system capable of converting surrounding renewable energy into additional electricity, thereby reducing the load on the main grid and minimising the risk of breaker trips.

VoltraBloom is a portable, hybrid renewable power system inspired by Smartflower, combining solar panels, a vertical-axis wind turbine, and a Soil Microbial Fuel Cell (SMFC). Integrated with LiFePO₄ batteries, and IoT monitoring, it serves as both a backup power solution for buildings with unstable grid supply and an independent energy source for diesel-dependent communities, many of which still rely heavily on diesel generators due to limited grid infrastructure (Saleh, 2026).

Since the concept was first formulated, the design team has proceeded to build a small physical proof-of-concept alongside the implementation-scale design presented in this paper. The proof-of-concept uses accessible materials and components — four selected solar panels, a DC-motor wind demonstrator, a simplified 2S storage path, and ESP32-based control — to verify that the intended architecture can be assembled and operated as one coordinated system before the full-scale ratings described below are pursued.

1.2 Problem Formulation

1. How can VoltraBloom be designed as a renewable-energy-based backup power source for dormitories, small buildings, and the wider community?
2. How can a portable floral structure be designed to house all system components?
3. How can an IoT-based monitoring system be designed to monitor VoltraBloom performance in real time?
4. How can the intended architecture be verified through a small physical proof-of-concept prior to implementation-scale development?

1.3 Research Objectives

1. To produce a conceptual design of VoltraBloom as a renewable-energy-based backup power system.
2. To design an IoT-based monitoring system for real-time performance monitoring.
3. To build and document a physical proof-of-concept prototype that verifies the source, storage, control, and mechanical architecture at a small scale.

1.4 Scope and Limitations

This study covers two development levels. The first is the implementation-scale conceptual design, which defines the intended architecture, component specifications, energy capacity, and operational concept of VoltraBloom as a future hybrid renewable emergency power system. The second is the physical proof-of-concept prototype, which uses simplified and accessible components to demonstrate selected mechanical, electrical, and control principles of the proposed architecture.

The physical prototype is not intended to reproduce the electrical capacity, structural dimensions, battery capacity, or performance of the implementation-scale system. Its primary function is to demonstrate that the proposed floral structure, photovoltaic arrangement, wind-energy representation, energy storage pathway, and basic control mechanism can be physically realised. Therefore, prototype observations are interpreted as proof-of-concept evidence rather than as full-scale performance results.

1.5 Research Benefits

Theoretical Benefits. Contribute references on the design of building-scale hybrid renewable energy systems that integrate soil microbial fuel cells, and IoT-based monitoring, together with a documented pathway from conceptual design to a verified physical proof of concept.

Practical Benefits. Provide an alternative portable, efficient, and environmentally friendly backup power system for boarding schools and the wider community. SMAN CMBBS will serve as the initial site for problem identification.

CHAPTER II

LITERATURE REVIEW

2.1 Theoretical Foundations

2.1.1 Basic Electrical Concepts

Electricity is the flow of charge utilised as an energy source (KBBI, 2016). VoltraBloom uses a 48 V DC system on the generation and storage side, which is then converted to 220 V AC via an inverter to supply building loads at the implementation scale; the physical proof-of-concept uses a lower, prototype-appropriate DC voltage rather than the full 48 V bus.

2.1.2 Renewable Energy

Renewable energy comes from natural sources that can be replenished and is more environmentally friendly than fossil energy (Mayasari et al., 2022). Its utilisation underpins the development of sustainable backup power systems.

2.1.3 Solar Panels

Solar panels convert sunlight into electricity through the photovoltaic effect (Mayasari et al., 2022). In VoltraBloom, solar panels are the primary energy source for charging the battery bank; the proof-of-concept demonstrates this principle using four selected panels chosen after basic light-response checks of nine available units.

2.1.4 Batteries

Batteries store electrical energy so it can be used when the primary energy sources are unavailable (Utomo et al., 2023). VoltraBloom's implementation-scale concept uses a 48 V 200 Ah LiFePO₄ battery bank because of its long service life and high safety; the proof-of-concept instead uses a simplified two-cell (2S) storage path pending verification of chemistry and specification.

2.1.5 Vertical-Axis Wind Turbine

Wind turbines convert the kinetic energy of wind into electricity. VoltraBloom employs a vertical-axis wind turbine as a complementary energy source to enhance system reliability, especially when solar irradiance is low (Wang et al., 2023). At prototype scale, a DC motor operated in reverse is used as a small generator demonstrator to validate the conversion principle.

2.1.6 Soil Microbial Fuel Cell

Soil Microbial Fuel Cells (SMFCs) generate electricity from the activity of electrogenic bacteria in soil (Simeon et al., 2022). Recent engineering guidelines for soil-powered computing indicate that individual SMFC units typically generate power in the milliwatt range, making them suitable for low-power applications such as environmental sensors rather than a primary power supply (Yen et al., 2024). In VoltraBloom, the Soil Microbial Fuel Cell (SMFC) functions as a low-power supplementary energy source to support sensors and the monitoring system, both at implementation scale and in the prototype.

2.1.7 Quantification Basis: Power, Energy, and Efficiency

To keep the Physics and Engineering contribution explicit and to support quantitative testing of the prototype (Section 3.7), this paper uses the following governing relations. Any numerical value derived from these relations is labelled either as a measured prototype result, or as an implementation-scale estimate, and the two are never merged in the same table without a label.

Electrical power: $P = V \times I$, where P is power in watts (W), V is voltage in volts (V), and I is current in amperes (A).

Energy over an interval: $E = \int P(t) dt$; for measurements taken at fixed intervals this is approximated as $E \approx \sum P_i \Delta t$.

Conversion efficiency: $\eta = (P_{\text{output}} / P_{\text{input}}) \times 100\%$.

Battery energy (first-order estimate): $E_{\text{battery}} \approx V_{\text{nominal}} \times Ah$, where the nominal voltage depends on the specific cell chemistry used; this value is not assumed for the prototype's 2S configuration until the cell type, capacity, and charge limits are confirmed.

These relations are applied to the raw voltage and current readings collected in the subsystem tests of Table 3.7 and to the implementation-scale component specifications of Table 4.1; results calculated from specifications are reported as design targets, and results calculated from prototype measurements are reported as observed values.

2.1.8 MPPT Charge Controller and Inverter

An MPPT charge controller optimises battery charging, while an inverter converts 48 V DC to 220 V AC for building loads at the implementation scale. Neither the MPPT charge controller nor mains-level inversion has yet been validated in the physical prototype.

2.1.9 IoT-Based Monitoring System

The IoT-based monitoring system uses sensors and an ESP32 microcontroller to monitor voltage, current, battery state-of-charge, irradiance, wind speed, and temperature in real time at implementation scale. This layer is treated as a future addition once the prototype's electrical and mechanical hardware has been verified.

2.1.10 VoltraBloom as a Hybrid System

VoltraBloom integrates solar panels, a wind turbine, a Soil Microbial Fuel Cell (SMFC), a battery bank, MPPT charge controllers, and IoT-based monitoring into a single efficient and sustainable backup power system (Kassem et al., 2024). This integrated approach aligns with recent developments in hybrid solar-wind systems designed for in-house power generation, which demonstrate improved reliability through source diversification (Alnaimi et al., 2024). The physical proof-of-concept realises the same architecture at a reduced scale to verify that the subsystems can be assembled and controlled as one coherent system.

2.2 Review of Previous Research

Previous studies have generally developed hybrid solar-wind systems for building energy needs. However, most of them have not integrated Soil Microbial Fuel Cells, foldable design, and IoT monitoring within one system. This gap forms the basis for developing VoltraBloom.

Aspect	VoltraBloom	Smartflower	Conventional EPK/Diesel Generator
Energy source	Solar+wind+ground	Solar	Solar/battery/Fossil/fuel
Power capacity	± 3.2 kWp + 800 W (wind), 5 kW AC output	± 3.2 kWp (solar only)	Varies, typically generator-dependent
Portability	High (foldable, wheeled)	Medium	Low
Emissions	None	None	High (Diesel Generator)
Estimates cost	Medium-high	High	Medium-high
Monitoring	Real-time IoT	Limited	Typically none

Table 2.1 Comparison of VoltraBloom with Existing Backup Power Solutions

2.3 Conceptual Framework

This research originates from limited electricity supply in boarding schools and dependence on diesel generators. Based on renewable energy, hybrid systems, HEMS,, and IoT monitoring, VoltraBloom is designed as a renewable backup power system at the implementation scale. The expected outputs are the system design, block diagram, flowchart, mechanical design, layout, wiring diagram, dimension drawing, exploded view, and component specifications. A physical proof-of-concept prototype, built with accessible materials and simplified components, verifies the same architecture prior to full-scale implementation and provides the evidence basis for further development.

CHAPTER III RESEARCH METHODS

3.1 Type and Research Approach

This study uses a Conceptual Design Research (Research by Design) approach. The focus is on designing a system to solve the research problem at the implementation scale, complemented by a physical proof-of-concept prototype used to verify the intended architecture without yet claiming full-scale performance.

3.2 Research Site and Period

The research was carried out by the SMAN CMBBS team from June to July 2026 as preparation for the Indonesia International Invention Expo (IIEX) 2026. Design activities were conducted digitally using design and online communication software, and the physical proof-of-concept was subsequently fabricated using accessible materials and components available to the team.

3.3 Research Stages

1. Literature review and problem identification.
2. System requirements analysis.
3. Overall system design, including block diagrams and flowcharts.
4. Mechanical design, component layout, and wiring diagram.
5. Preparation of component specifications and technical drawings.
6. Concept analysis and design evaluation based on literature and technical specifications.
7. Fabrication of a physical proof-of-concept prototype and documentation of component-level observations.

3.4 Data Collection Techniques

The research data consist of secondary data obtained from scientific literature, component datasheets, and technical references used to support the implementation-scale system design. Prototype-level data consist of component-selection records, fabrication observations, mechanical observations, and basic functional observations of the physical prototype.

At the current stage, the prototype has not yet produced a controlled quantitative dataset covering voltage, current, power output, energy yield, charging efficiency, battery runtime, or long-term operating performance. Therefore, numerical performance values presented for the implementation-scale system are treated as calculated estimates or design targets and are not presented as experimental results from the physical prototype.

3.5 Design Tools

The design process used Draw.io for diagrams, Canva for conceptual illustrations, and SketchUp for 3D visualisation. The physical proof-of-concept was fabricated using PVC pipe, lightweight board/triplek, hinges, and an Arduino microcontroller programmed in the Arduino IDE.

3.6 Concept Analysis Techniques

The analysis was conducted by evaluating design conformity with the research problem, comparing it with similar technologies, and assessing feasibility based on component capacities and specifications.

3.7 Physical Proof-of-Concept Development

To verify that the design architecture is physically realisable, the team decomposed the system into source, conditioning/storage, mechanical, control, and switching subsystems and fabricated a small-scale demonstrator. Each subsystem was intended to be checked individually before integrated testing, following the stages below.

Subsistem	Pengujian yang Direncanakan	Data / Kriteria Verifikasi
Panel surya	Diukur di bawah kondisi cahaya yang sama menggunakan multimeter dan sensor arus	V open-circuit, V berbeban, I, dan P per panel
Generator angin (demonstrator motor DC)	Diputar pada beberapa tingkat kecepatan mekanis	Tegangan, arus, dan daya keluaran pada tiap tingkat kecepatan
Soil Microbial Fuel Cell (SMFC)	Diukur satu pot dan gabungan paralel, tanpa beban dan dengan beban	V open-circuit, V berbeban, I, dan P saat diberi beban
Baterai (konfigurasi 2S)	Pengisian dan pelepasan terkendali menggunakan multimeter/logger	Tegangan, arus, dan durasi pengisian/pengosongan
Kontrol (ESP32)	Pembacaan sensor dan status logika kontrol diuji secara terpisah	Nilai sensor dan kestabilan pembacaan
Mekanik (buka-tutup kelopak)	Diuji buka-tutup berulang menggunakan servo aktuator	Waktu siklus, tingkat keberhasilan, dan kestabilan struktur

Table 3.7 Physical proof-of-concept development stages

The distinction between observed, calculated, specified, and targeted values is maintained throughout this paper: a value is presented as a prototype result only if it has

actually been observed or measured on the physical prototype; otherwise it is identified as an implementation-scale target.

CHAPTER IV RESULTS AND DISCUSSION

4.1 Overall System Design

VoltraBloom is designed to resemble a flower, featuring four solar panel petals, a vertical-axis wind turbine at the top, a telescopic pole, and a base that houses all electronic components and caster wheels for mobility. In open mode, the petals unfold to maximise sunlight absorption, while in closed mode they fold inward to make the system more compact and portable. This is the implementation-scale concept; Section 4.10 presents the equivalent architecture as realised in the current five-petal physical proof-of-concept.

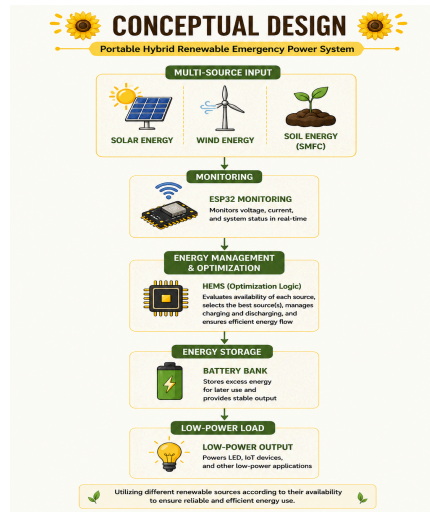


Figure 4.1: VoltraBloom Conceptual Design

4.2 System Block Diagram

VoltraBloom uses three energy sources: solar panels, vertical-axis wind turbines, and a soil module. The harvested energy is regulated by a charge controller and managed by the HEMS before being stored in the battery bank. The inverter then converts DC into 220 V AC to supply building loads, and the entire process is monitored through an IoT-based system. This block diagram reflects the implementation-scale target; the proof-of-concept realises the same energy-and-control logic using simplified conditioning, 2S storage, and Arduino-based control rather than a full MPPT/HEMS/IoT stack (see Figure 4.6).

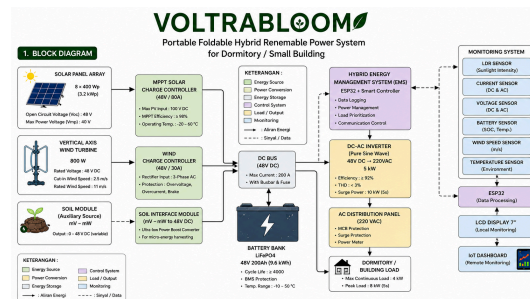


Figure 4.2: VoltraBloom System Block Diagram

4.3 Component Specification (Implementation Scale)

No	Component	Specification	Amount
1	Solar Panel	400 Wp	8 units
2	Wind Turbine	800 W	1 unit
3	MPPT Solar Charge Controller	48V/80A	1 unit
4	AC Rectifer	48 V DC output	1 unit
5	Battery Bank (LifePO4)	48V 200Ah (9,6 kWh)	1 unit
7	Hybrid EMS Controller	ESP32+Smart Controller	1 set
8	AC Distribution Panel	220VAC	1 unit
9	DC Breaker	63A	2 unit
10	AC Breaker (MCB)	32A	1 unit
11	LCD Display	7 inci TFT	1 unit
12	Caster Wheel (Lockable)	Heavy Duty	4 unit
13	Main Cabinet Material	Alumunium+steel	1 set
14	Pole Material	Aluminium Telescopic	1 unit
15	Protection	Fuse, SPD, Grounding	1 set

Table 4.1: VoltraBloom Component Specification (Implementation-Scale Target)

4.4 Operational Flowchart

The system starts when the solar panels and wind turbines generate energy. The HEMS then manages battery charging and selects the distribution path based on system conditions. Energy is delivered either to DC loads or converted to AC through the inverter, and all parameters are monitored in real time. At prototype scale, the equivalent control sequence (initialise → read input → decide state → actuate servo/relay → repeat, with a fault state defaulting to safe-off) is implemented on the Arduino, as shown in Figure 4.7.

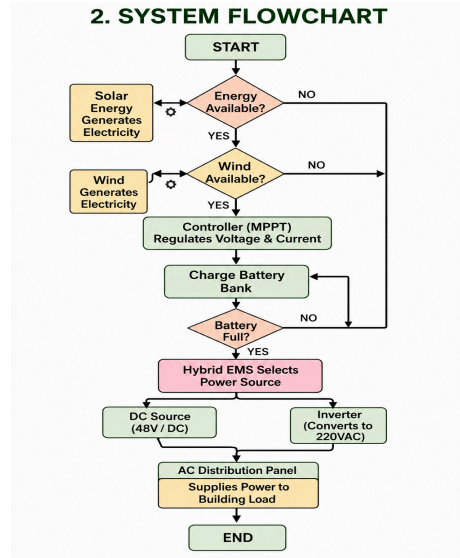


Figure 4.3: VoltraBloom Operational Flowchart

Note: This figure represents the proposed implementation-scale design and is not a photograph or measurement result from the current physical prototype.

4.5 Mechanical Design and Dimension Drawing

The mechanical design is shown in front, side, top. The foldable floral structure makes the system efficient and portable. Based on typical component weights, the estimated total system weight is approximately 250–300 kg. The physical proof-of-concept instead uses a lightweight PVC-and-triplek structure sized for five panels, described in Section 4.10.

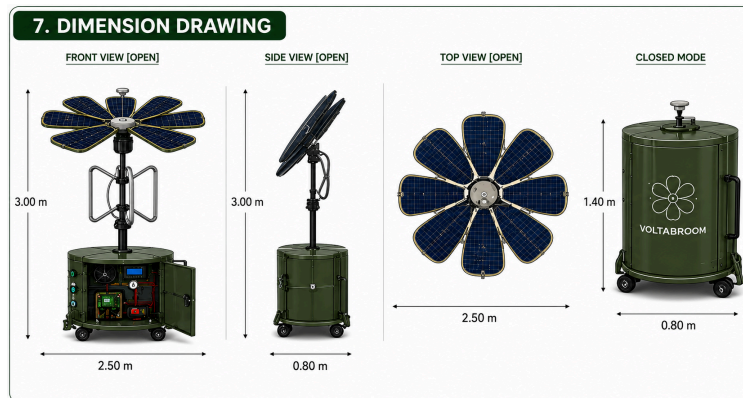


Figure 4.4: VoltraBloom Dimension Drawing (Front, Side, Top View, dan Closed Mode)

Note: This figure represents the proposed implementation-scale design and is not a photograph or measurement result from the current physical prototype.

4.6 Component Layout

At implementation scale, all major components, including the battery bank, charge controller, HEMS, distribution panel, and LCD, are housed in the base. This layout supports easier installation, maintenance, and cable routing. In the proof-of-concept, the equivalent low-voltage components (2S storage and Arduino) are housed at the base of the PVC stem.



Figure 4.5: VoltraBloom Component Layout inside the VoltraBloom Base

Note: This figure represents the proposed implementation-scale design and is not a photograph or measurement result from the current physical prototype.

4.7 Electrical Wiring Diagram

The implementation-scale wiring diagram shows the electrical connections between the energy sources, charge controller, battery bank, inverter, distribution panel, and sensors, with power lines and data lines separated to improve safety and simplify maintenance. Figure 4.6 in Section 4.10 shows the corresponding system-level wiring concept used in the physical proof-of-concept.

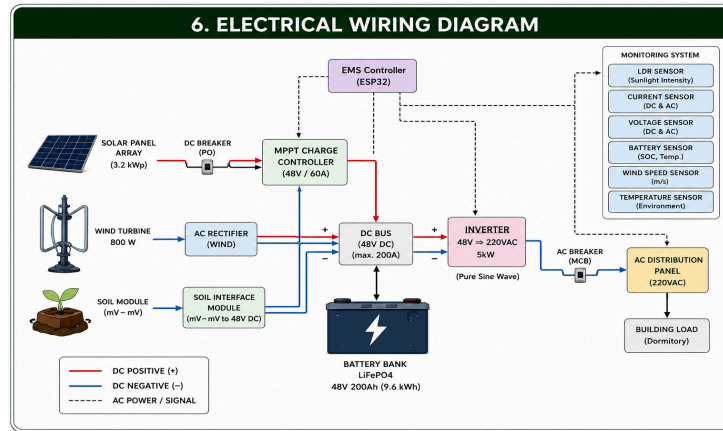


Figure 4.6: VoltraBloom Electrical Wiring Diagram

Note: This figure represents the proposed implementation-scale design and is not a photograph or measurement result from the current physical prototype.

4.8 Analysis of Concept and Implementation Potential

The design indicates that VoltraBloom has strong potential as an efficient, environmentally friendly backup power system, integrating solar panels, wind turbines, and a soil module, with HEMS enabling automatic energy management and the foldable design improving portability. Compared with similar systems, VoltraBloom offers greater energy-source diversity, portability, and integrated real-time monitoring. Economically, against a typical diesel generator's estimated fuel cost of IDR 1,500,000–2,500,000/month, VoltraBloom's total investment of approximately IDR 98,100,000 (USD 6,131) could theoretically reach cost parity within 3–5 years, after which it would operate at near-zero marginal cost with no direct emissions, though this estimate excludes maintenance costs and requires further validation. The physical proof-of-concept described in Section 4.10 is the first step toward that validation: it demonstrates that the source, storage, control, and mechanical subsystems can be integrated and operated together, ahead of any economic or full-scale performance claim.

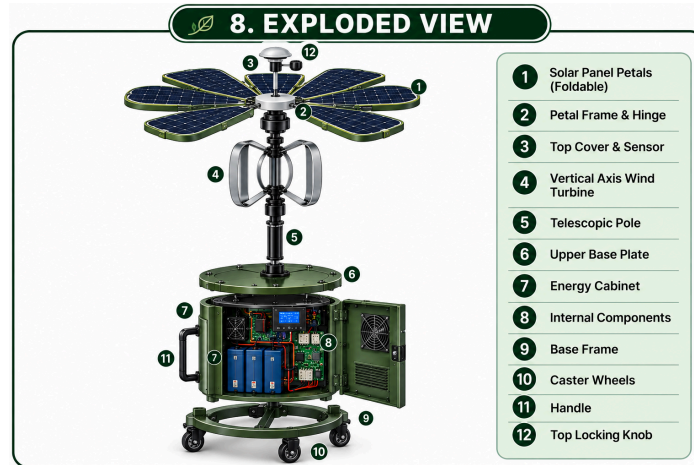


Figure 4.7: VoltraBloom Exploded View

Note: This figure represents the proposed implementation-scale design and is not a photograph or measurement result from the current physical prototype.

4.9 Capacity and Performance Estimation (Implementation Scale)

Based on component specifications and reference literature, the solar array serves as the primary energy source and the wind turbine as a supplementary one. Assuming 4 peak sun hours/day and an 80% derating factor, the 3.2 kWp solar array is estimated to yield approximately 10.2 kWh/day; the 800 W wind turbine, at a 15% capacity factor, contributes a further 2.9 kWh/day, for a combined estimated harvest of ~13 kWh/day. The SMFC module produces only milliwatt-to-watt-level output (Yen et al., 2024) and is allocated solely to low-power sensors and monitoring rather than the main load. The 9.6 kWh LiFePO₄ battery bank, at 80% DoD, provides ~7.7 kWh of usable capacity — sufficient for approximately 7–8 hours of backup autonomy at a 1 kW critical dormitory load without concurrent generation, and considerably longer with daytime charging. These figures remain theoretical estimations based on standard literature assumptions and require validation through prototype testing; they have not yet been measured on the physical proof-of-concept.

No.	Component	Estimated Unit Price (IDR)	Estimated Unit Price (USD)	Qty	Estimated Subtotal (IDR)	Estimated Subtotal (USD)
1	Solar Panel Monocrystalline 400 Wp	Rp2,200,000	\$138	8	Rp17,600,000	\$1,100
2	Wind Turbine, Vertical Axis, 800 W	Rp9,000,000	\$563	1	Rp9,000,000	\$563
3	MPPT Charge Controller 48V/80A	Rp3,500,000	\$219	1	Rp3,500,000	\$219
4	Battery Bank LiFePO4, 48V 200Ah	Rp30,000,000	\$1,875	1	Rp30,000,000	\$1,875
5	Inverter, Pure Sine Wave, 5 kW	Rp17,000,000	\$1,063	1	Rp17,000,000	\$1,063
6	Hybrid EMS Controller (ESP32 + Relay)	Rp1,500,000	\$94	1	Rp1,500,000	\$94
7	IoT Monitoring Sensor Set	Rp1,500,000	\$94	1	Rp1,500,000	\$94
8	Breaker, Fuse, SPD, Grounding	Rp3,000,000	\$188	1	Rp3,000,000	\$188
9	Frame, Pole, Base, Wheel, Hinge	Rp12,000,000	\$750	1	Rp12,000,000	\$750
10	Cable, Connector, etc.	Rp3,000,000	\$188	1	Rp3,000,000	\$188
Total					Rp98,100,000	\$6,131

**USD values are approximate conversions using an assumed exchange rate of IDR 16,000 = USD 1 for reference purposes only; actual costs will vary with market prices and prevailing exchange rates at time of implementation.*

Table 4.2: VoltraBloom Estimated Cost Feasibility of VoltraBloom Components (Implementation Scale)

4.10 Physical Proof-of-Concept Prototype

To verify the architecture described in Sections 4.1–4.9 before implementation-scale fabrication, the team built a small physical proof-of-concept. Four solar panels were selected after nine available panels were individually checked for basic response to light. A DC motor is used as a small wind-generator demonstrator, while the SMFC is treated as a low-power supplementary source. The prototype employs a simplified 2S storage configuration, an Arduino controller programmed in the Arduino IDE. The mechanical structure uses accessible materials including PVC pipe, lightweight board/triplek, hinges, and a simple linkage.

4.10.1 What the Prototype Currently Demonstrates

The project record supports three qualitative, component-level observations: the available solar panels responded to light and four were selected for the prototype; the available DC motor/turbine arrangement can demonstrate wind-to-electric conversion when mechanically driven; and the proposed mechanical/control architecture is feasible as a proof-of-concept

assembly using PVC, board/triplek, and hinges. These are feasibility observations rather than measured performance results, and they are not converted into numerical efficiency, power, runtime, or energy-yield claims until a controlled measurement dataset is recorded.

Aspek	Skala Implementasi (Target Desain)	Prototipe Saat Ini (Proof of Concept)
Solar	Target 8 x 400 Wp ($\approx 3,2$ kWp)	4 panel terpilih dari 9 unit yang diuji respons cahaya, disusun dalam 2 rangkaian seri
Wind	Vertical-axis wind turbine 800 W	Motor DC dioperasikan sebagai demonstrator generator angin
SMFC	Modul tambahan berdaya rendah untuk sensor/monitoring	Pot kecil (elektroda carbon felt), dirangkai paralel; pembacaan awal $\approx 0,030$ V (kondisi pengukuran perlu dikonfirmasi: open-circuit/berbeban, satu pot/gabungan)
Storage	48V 200Ah LiFePO4 ($\approx 9,6$ kWh)	Li-ion 18650, konfigurasi 2S, dengan holder baterai
Controller	HEMS berbasis ESP32 + smart controller	ESP32, diprogram melalui Arduino IDE
Mekanik/Aktuasi	Kelopak 8 panel, rangka aluminium+baja, estimasi 250-300 kg	Struktur PVC dan triplek, aktuator servo, relai dua kanal

Table 4.10.1 Comparison of Implementation-Scale Target and Current Prototype

4.10.2 What the Prototype Currently Demonstrates

The project record supports three qualitative, component-level observations: the available solar panels responded to light and five were selected for the prototype; the available DC motor/turbine arrangement can demonstrate wind-to-electric conversion when mechanically driven; and the proposed mechanical/control architecture is feasible as a proof-of-concept assembly using PVC, board/triplek, hinges, servo actuation, and ESP32-based control. These are feasibility observations rather than measured performance results, and they are not converted into numerical efficiency, power, runtime, or energy-yield claims until a controlled measurement dataset is recorded.

4.10.3 What the Prototype Does Not Yet Demonstrate

Aspek yang Belum Dibuktikan	Alasan / Data yang Masih Diperlukan
Kapasitas dan performa listrik skala implementasi (3,2 kWp; 800 W; 5 kW)	Belum ada pengukuran V, I, dan P terkontrol pada prototipe; angka ini adalah target desain berdasarkan spesifikasi komponen
Efisiensi konversi masing-masing sumber energi (η)	Memerlukan pengukuran P input dan P output secara bersamaan; belum dilakukan
Kapasitas dan lama waktu backup baterai	Prototipe baru menggunakan sel 2S sederhana; konfigurasi 48V 200Ah LiFePO4 belum difabrikasi maupun diuji
Kinerja MPPT, inverter, dan HEMS skala penuh	Komponen ini belum dipasang pada prototipe; prototipe menggunakan Arduino dan relai dua kanal sebagai gantinya
Keluaran daya SMFC dalam kondisi nyata	Baru diketahui dari literatur (rentang mikrowatt-miliwatt); belum diukur langsung pada pot SMFC prototipe
Pemantauan berkala jangka panjang (6-24 jam)	Belum dilakukan; memerlukan interval pencatatan, kondisi lingkungan, dan sensor yang terdefinisi
Validasi ekonomi dan payback period	Estimasi Rp98.100.000 belum memperhitungkan biaya perawatan dan belum divalidasi di lapangan

The most defensible contribution of the current prototype is integration: multiple renewable inputs are represented in one architecture, the flower structure provides a physical energy-collection form, and the control layer connects electrical conditions to mechanical and switching actions. The next research value comes from quantitative measurement of the prototype, followed by an evidence-based scale-up toward the implementation-scale targets described in Sections 4.1–4.9.

CHAPTER V

CONCLUSION

VoltraBloom is a conceptual design of a hybrid renewable power system at an implementable scale, intended as a backup power source for dormitories or small buildings when the main electricity supply is interrupted or insufficient. The system integrates solar panels, a vertical-axis wind turbine, and a soil microbial fuel cell as energy sources, then regulates the generated power through an MPPT charge controller, battery bank, HEMS, and IoT-based monitoring system.

Since the original conceptual design was formulated, the team has built and documented a small physical proof-of-concept that realises the same source – conditioning/storage – control – mechanical architecture at a reduced scale, using four selected solar panels, a DC-motor wind demonstrator, a 2S storage path, and ESP32-based control. This prototype confirms that the intended architecture can be physically assembled and operated as one coordinated system, without yet establishing the electrical performance, capacity, or runtime figures associated with the implementation-scale concept.

Based on the conceptual analysis, VoltraBloom has the potential to become a more environmentally friendly backup power alternative than diesel generators and can support basic electrical loads in dormitories or small buildings. However, because the implementation-scale system remains at the conceptual stage and the physical prototype has not yet undergone controlled quantitative testing, its performance and efficiency must be validated through further prototype measurement, refinement, and scale-up before implementation-scale claims can be made.

REFERENCES

- Alnaimi, F. B. I., Kazem, H. A., Alzakri, A. B., & Alatir, A. M. (2024). Design and implementation of smart integrated hybrid solar-Darrieus wind turbine system for in-house power generation. *Renewable Energy and Environmental Sustainability*, 9, 2.
- Jamaaluddin, J., Ayuni, S. D., & Wulandari, I. A. S. (2023). Design of automatic transfer switch system solar power plant - PLN. *Journal of Electrical Technology UMY*, 7(2).
- Kassem, Y., Camur, H., & Ghoshouni, E. G. (2024). Assessment of a hybrid (wind-solar) system at high-altitude agriculture regions for achieving Sustainable Development Goals. *Engineering, Technology & Applied Science Research*, 14(1), 12595-12607.
- KBBI. (2016). *Kamus Besar Bahasa Indonesia*. Jakarta: Badan Pengembangan dan Pembinaan Bahasa.
- Manaf, A. (2022). Rekonstruksi pendidikan boarding school di Indonesia. *Jurnal Ad-Dawah*, 20(1).
- Mayasari, F., Samman, F. A., Muslimin, Z., Waris, T., Dewiani, Salam, A. E. U., Gunadin, I. C., Areni, I. S., Akil, Y. S., & Rachmaniar, I. (2022). Pengenalan panel surya sebagai salah satu sumber energi terbarukan. *Jurnal Tepat (Teknologi Terapan untuk Pengabdian Masyarakat)*, 5(2).
- Saleh, M. (2026, June 5). Potensi besar energi terbarukan desa sia-sia: 658 ribu keluarga hidup tanpa listrik, jutaan warga masih andalkan diesel. *The Conversation*. Retrieved from <https://theconversation.com>
- Saputro, J. S., dkk. (2023). Energy monitoring and control of automatic transfer switch between grid and solar panel for home system. 3(1), 59-73.
- Simeon, I. M., Weig, A., & Freitag, R. (2022). Optimisation of soil microbial fuel cell for sustainable bio-electricity production. *Biotechnology for Biofuels and Bioproducts*, 15, 124.
- U.S. Energy Information Administration. (2025, July 8). Electricity explained. Retrieved from <https://www.eia.gov/energyexplained/electricity/>
- Utomo, M. S., Nugrahanto, I., & Sungkono. (2023). Sistem penyimpanan energi menggunakan baterai sel sekunder pada photovoltaic. *Jurnal Elkolind*, 10(1), 85-92.

- Wang, H., Xiong, B., Zhang, Z., Zhang, H., & Azam, A. (2023). Small wind turbines and their potential for internet of things applications. *iScience*, 26(9), 107674.
- Yen, B., Jaliff, L., Gutierrez, L., Sahinidis, P., Bernstein, S., Madden, J., Taylor, S., Josephson, C., Pannuto, P., Shuai, W., Wells, G., Arora, N., & Hester, J. (2024).
- Soil-powered computing: The engineer's guide to practical soil microbial fuel cell design. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 7(4), 1.